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Modern Workplace Management Through a DevOps Lens

From Portal Clicking to Platform Engineering

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Microsoft

baseVISION
SECURE & MODERN ENDPOINT MANAGEMENT

EPIC FUSION
BRING IT ALL TOGETHER

Why are we still clicking?

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#WPNinjaCH

Focus

Cloud Engineer
Development and Automation Workplace Services

From

glueckkanja

My Blog

-

Certifications

AI-102 / MD-102 / MS-102 / MS-700

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Sport / E-Guitar / Photography / FPV-Drone / Motorbike / the list goes on…

Contact

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Agenda

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- What is DevOps really?
- Where is it used?
- The SCCM era
- Why MWP didn’t adopt DevOps
- The DevOps era

Key takeaways:

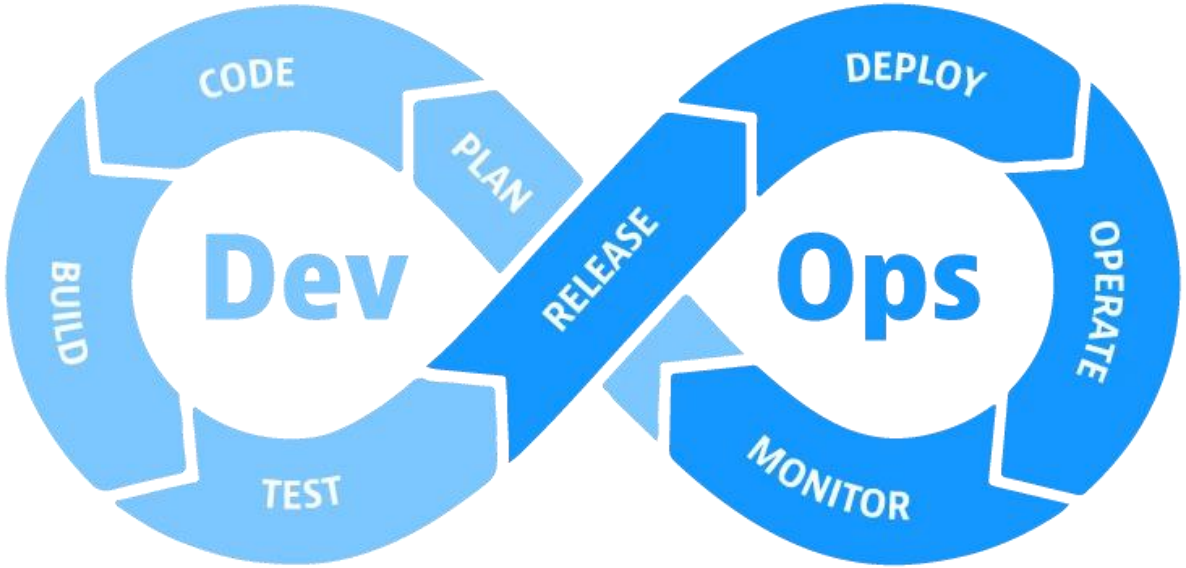
- Discover what automation can achieve
- Explore the tools behind it
- See how it works in practice

Agenda

- From clicks to code – With the right tools
- Demo time! ☺
- Q&A

What is DevOps really?

Development Operations



- **CI: Continuous Integration**
- **CD: Continuous Delivery**

"

DevOps is a mindset where teams define desired state in code and let automation enforce it - so they can move faster, reduce errors, and improve consistency.

"

Where is DevOps used?

Development Operations

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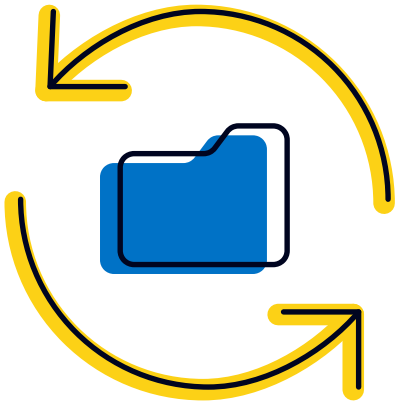


Infrastructure

- Ansible
- Bicep
- Terraform

ML & AI

- TensorFlow
- MLflow
- Kubeflow

Security

- SonarQube
- Snyk
- HashiCorp Vault

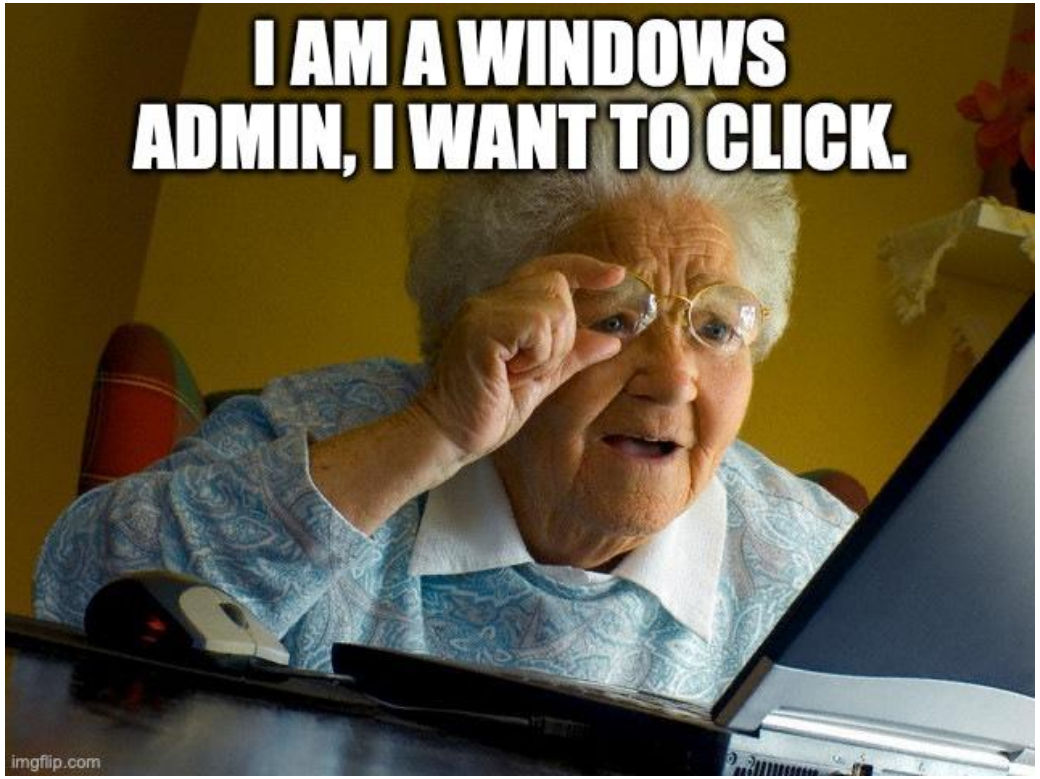
SRE

- Prometheus
- Grafana
- PagerDuty

The SCCM Era

When clicking was the norm

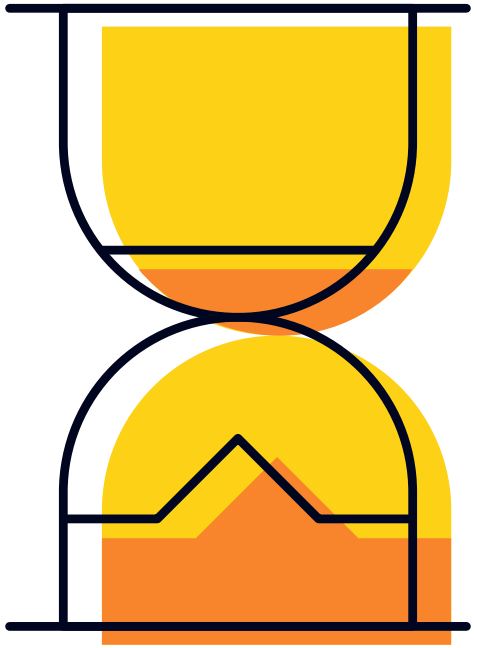
- **Manual Workflows**
  - Configuration done in GUI
  - Reliance on Documentation and Checklists
  - Slow deployments
- **Tooling Limitations**
  - No API, only Scripts
  - Built for On Prem Environments
- **Pain Points**
  - High Risk of Misconfiguration
  - Difficult to Audit or Replicate
  - Drift Between State and Documentation



Why Modern Workplace Didn’t Adopt DevOps

Why are we still clicking?

- Intune was built as a GUI-First Admin Portal (with SCCM in mind)
- ClickOps is easier then DevOps
- Auditing, Backing-Up and other features, came as an afterthought
- Automation requires a foreign toolchain
- There is (was) no native translation layer



The DevOps Era

What Problem are we trying to solve here?

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- Microsoft 365 (Intune and Entra) configuration is a complex beast
- If you are alone and have just one tenant, you may be ok with 'clicks in portals', but ...
- If you are a team of admins, have multiple tenants (staging/production | MSP) to manage,
  'clicks in portals' is not a scalable option.

#Buzzwords

- documentation
- versioning / release management
- 4-eyes principle (= review & approval)
- drift detection
- recovery
- scaling / tiering / automation

Automation is on the Horizon…

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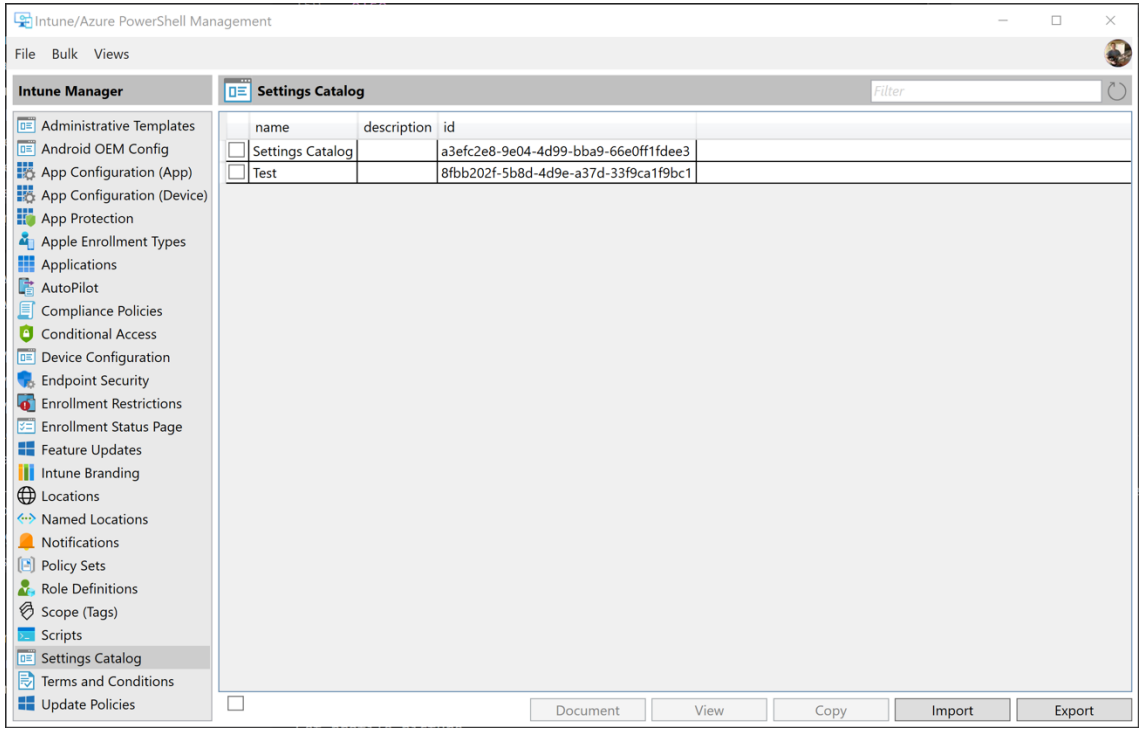
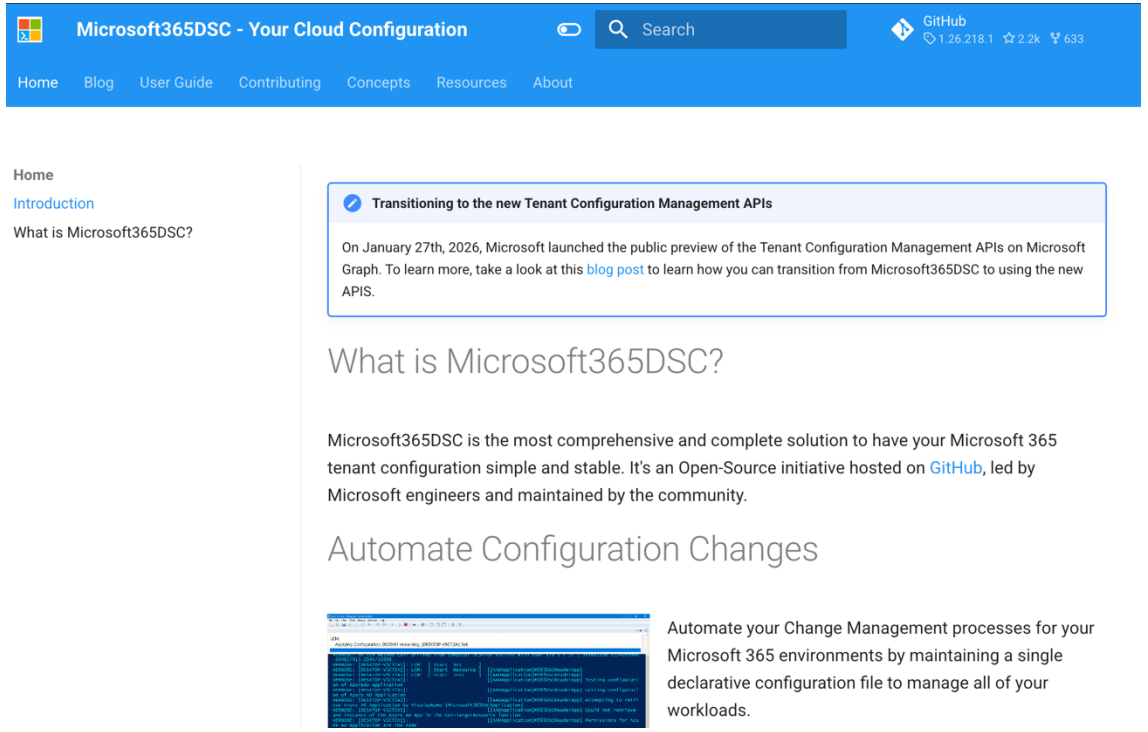
Micke-K/IntuneManagement

- Import and Export Intune Configuration
- Powered by Powershell



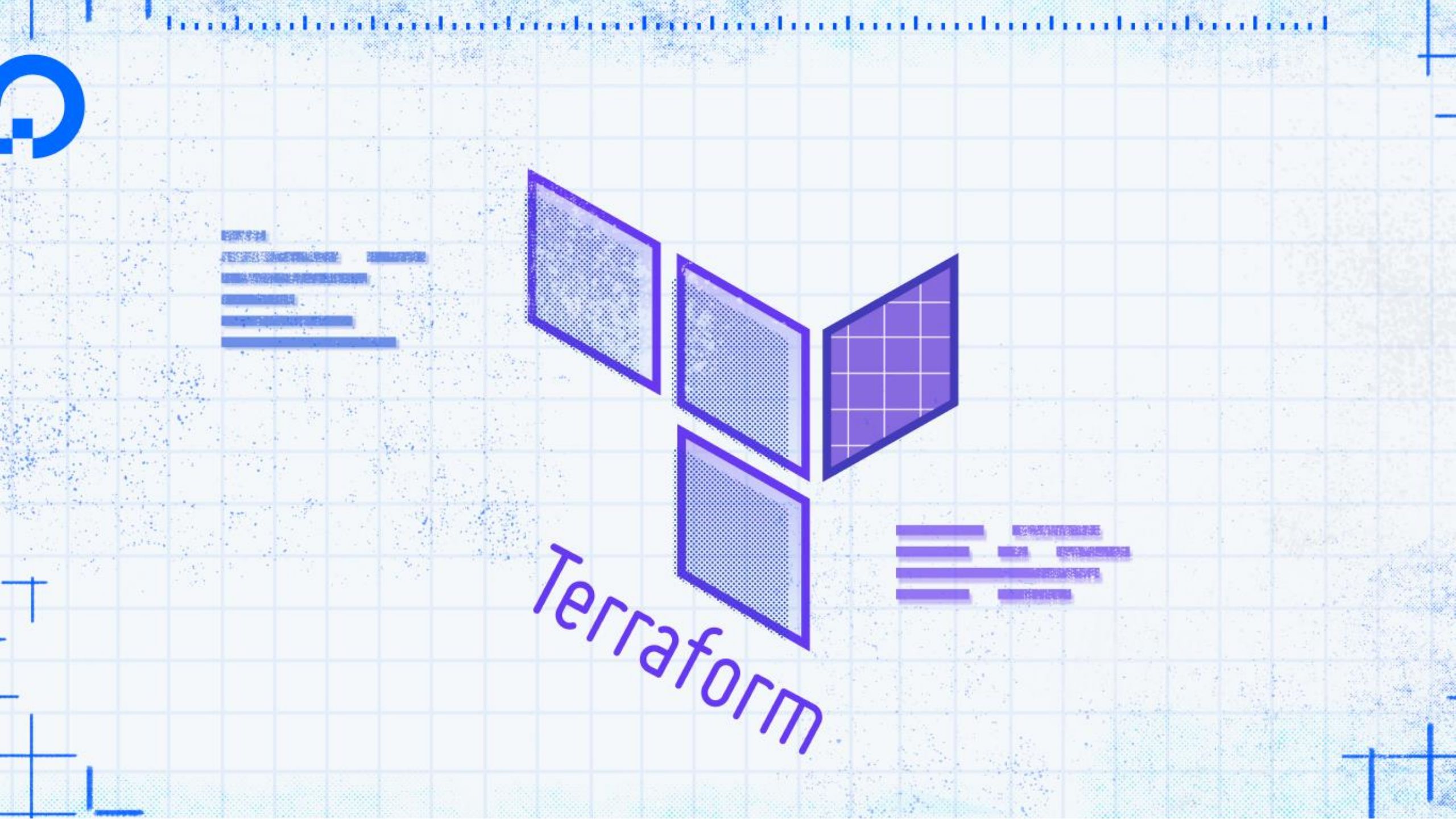
Microsoft365DSC

- Snapshots, Monitoring, Compare, etc.
- Powered by Powershell



Powershell? Nails everywhere…

- The **law of the instrument,** law of the hammer, ... is a cognitive bias that involves an over-reliance on a familiar tool. Abraham Maslow wrote in 1966, **"If the only tool you have is a hammer, it is tempting to treat everything as if it were a nail.“**
- Don't get me wrong:
  I ❤ Powershell – but is it the right weapon for **Configuration-as-Code**?
- Beside a lot of design considerations and caveats ...
  why not ignoring the hammer and check out which toolchain is used for **nearly 40% of worldwide cloud automation?**


Terraform

From Clicks to Code - With the right Tools

Terraform

Definition:

- Terraform is an Infrastructure as Code (IaC) tool
- Defines and manages infrastructure using declarative configuration files

Key Benefit:

- Automated, repeatable deployments across environments
- Version-controlled infrastructure in Code
- Reduced manual configuration and human error

Terraform – How it works

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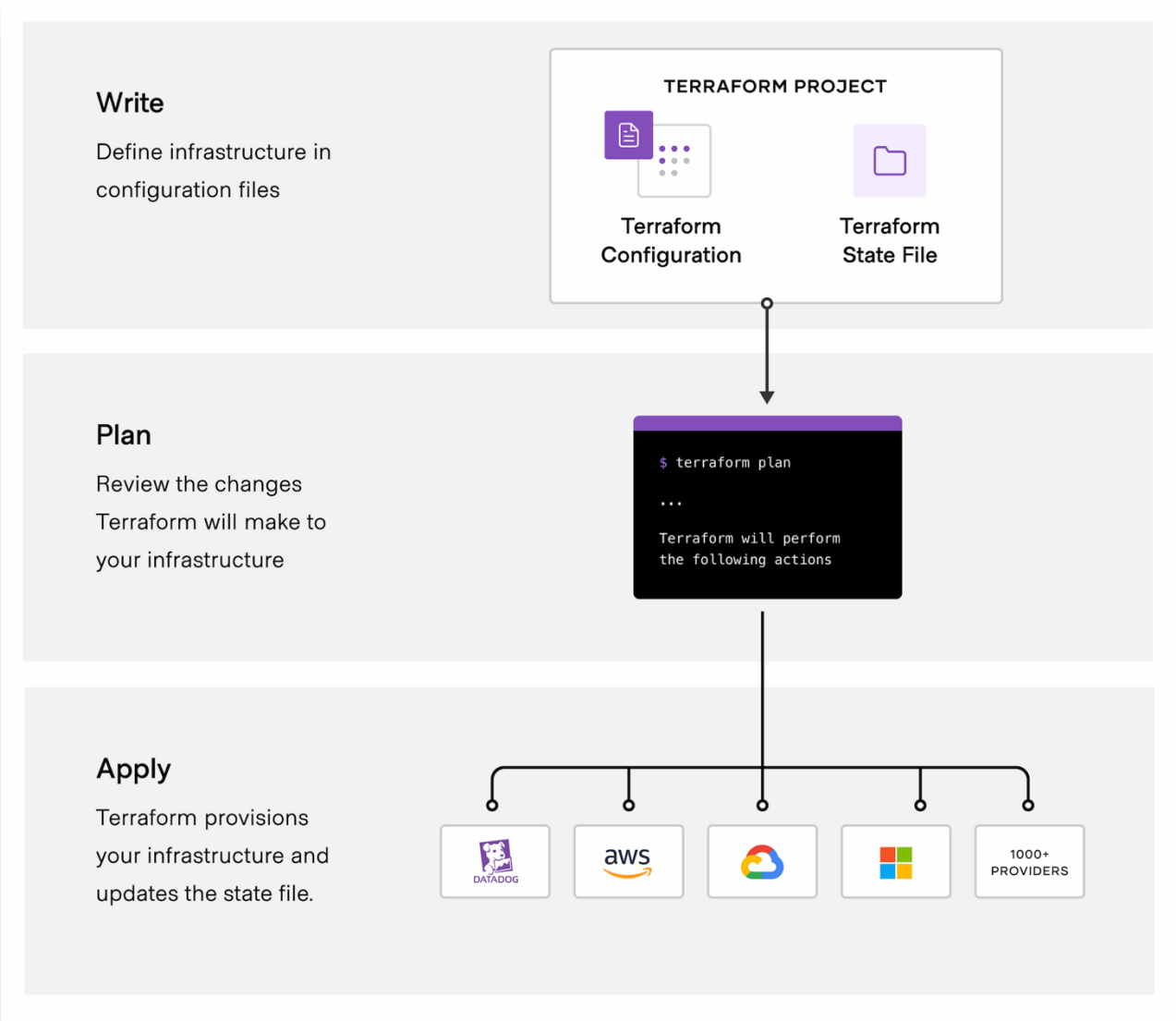
- **Write:** Infrastructure is written in configuration files
- **Plan:** Compares desired state vs. current state
- **Apply:** Applies approved changes to reach the desired state



The Important Link - Terraform Provider

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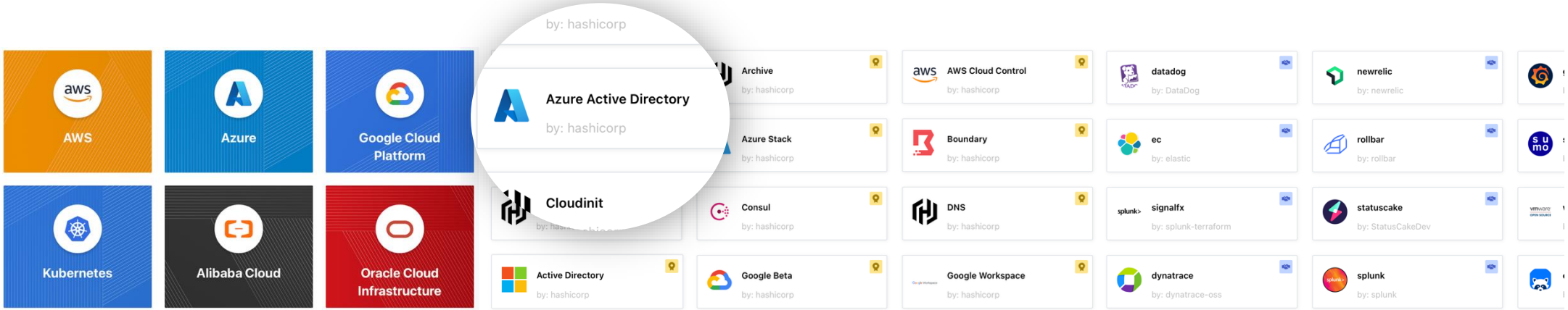

NEW: MSGraph Provider by Microsoft

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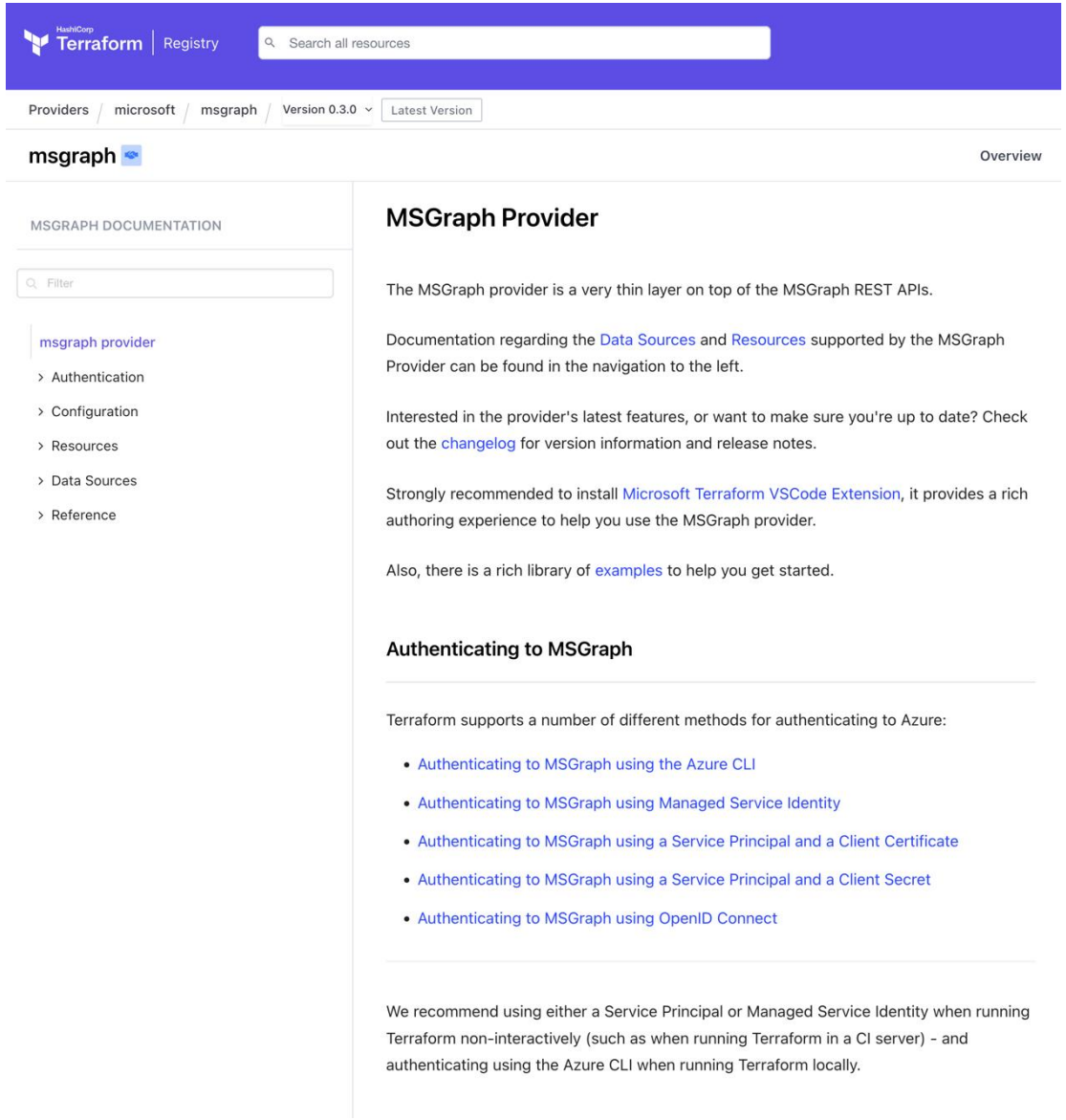
- The MSGraph provider is a very thin layer on top of the MSGraph REST APIs.
- Supported Ressources:
  - Applications
  - App role assignenemts
  - Federated identity credentials
  - Groups
  - OAuth2 permission grants
  - Service principals
  - Users



HashiCorp Terraform | Registry

Search all resources

Providers / microsoft / msgraph / Version 0.3.0 Latest Version

msgraph

Overview

MSGRAPH DOCUMENTATION

Filter

msgraph provider

- Authentication
- Configuration
- Resources
- Data Sources
- Reference

MSGraph Provider

The MSGraph provider is a very thin layer on top of the MSGraph REST APIs.

Documentation regarding the Data Sources and Resources supported by the MSGraph Provider can be found in the navigation to the left.

Interested in the provider's latest features, or want to make sure you're up to date? Check out the changelog for version information and release notes.

Strongly recommended to install Microsoft Terraform VSCode Extension, it provides a rich authoring experience to help you use the MSGraph provider.

Also, there is a rich library of examples to help you get started.

Authenticating to MSGraph

Terraform supports a number of different methods for authenticating to Azure:

- Authenticating to MSGraph using the Azure CLI
- Authenticating to MSGraph using Managed Service Identity
- Authenticating to MSGraph using a Service Principal and a Client Certificate
- Authenticating to MSGraph using a Service Principal and a Client Secret
- Authenticating to MSGraph using OpenID Connect

We recommend using either a Service Principal or Managed Service Identity when running Terraform non-interactively (such as when running Terraform in a CI server) - and authenticating using the Azure CLI when running Terraform locally.

TerraProvider

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- The Terraprovider is a translation layer with embedded Logic on top of the MSGraph REST APIs.
- Supported Ressources:
  - Assignements
  - Cloud PCs (W365 Provisioning Policies)
  - Corporate Deployment (Autopilot Profiles)
  - Device Management (Config Profiles/Scripts/Compliance)
  - Mobile App Management
  - Onboarding (Enrollment Configuration)
  - Software Updates
  - The List Goes On…

TerraProvider

So… where does it run?

Github

- Hosts your Code
- Git Code-Versioning
- 4-eyes Principle
- Release Management
- Issue Management

Github Actions

- Deploys Runners automatically
- Runs the Plan and Apply

- Demo

Q&A

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Thank you